

MIGRATION PLAN

Laptop → Cloud.

Moving TradePilot from a laptop in a bag to always-on infrastructure

**"Yahoo rate-limits, sleep bugs, and manual morning starts are symptoms of the same disease:
the wrong host."**

5

Phase plan

7

Days end-to-end

₹860

Monthly cost

6

Pain points solved

*TradePilot cloud migration · authored 2026-04-20 · target
Go-Live 2026-04-27*

Why Now

Author	Soumya Swain · soumya@sidewall.in
Trigger	Yahoo Finance 429 rate limits blocked Monday morning retrain. Second infrastructure failure in 7 days.
Goal	Deterministic daily runs, no manual pre-flight, no sleep bugs
Recommendation	DigitalOcean Bangalore droplet + Shoonya broker API + systemd services
Timeline	1 week from decision, parallel-run cutover (zero-risk)
Monthly cost	₹860 VPS + ₹0 Shoonya (free with trading account) = ₹860/month

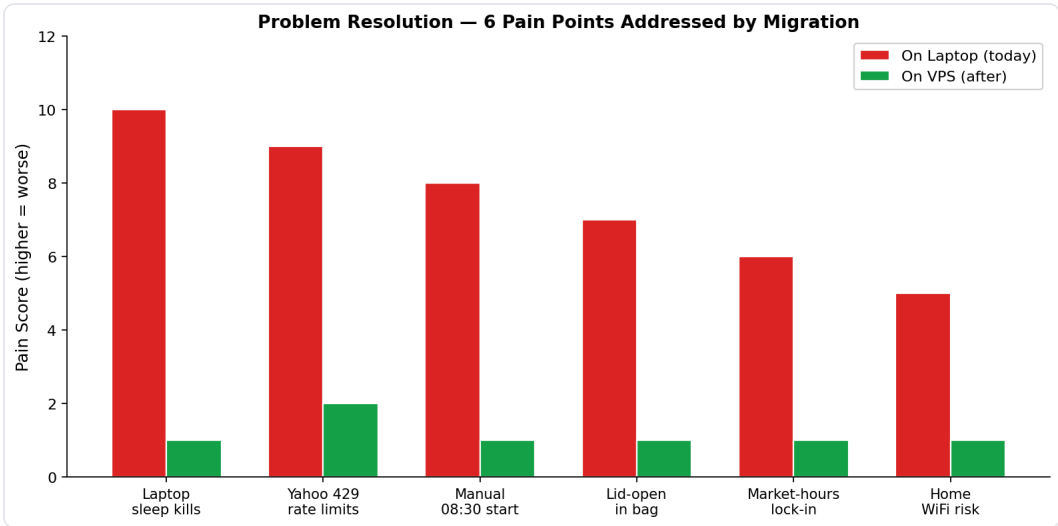
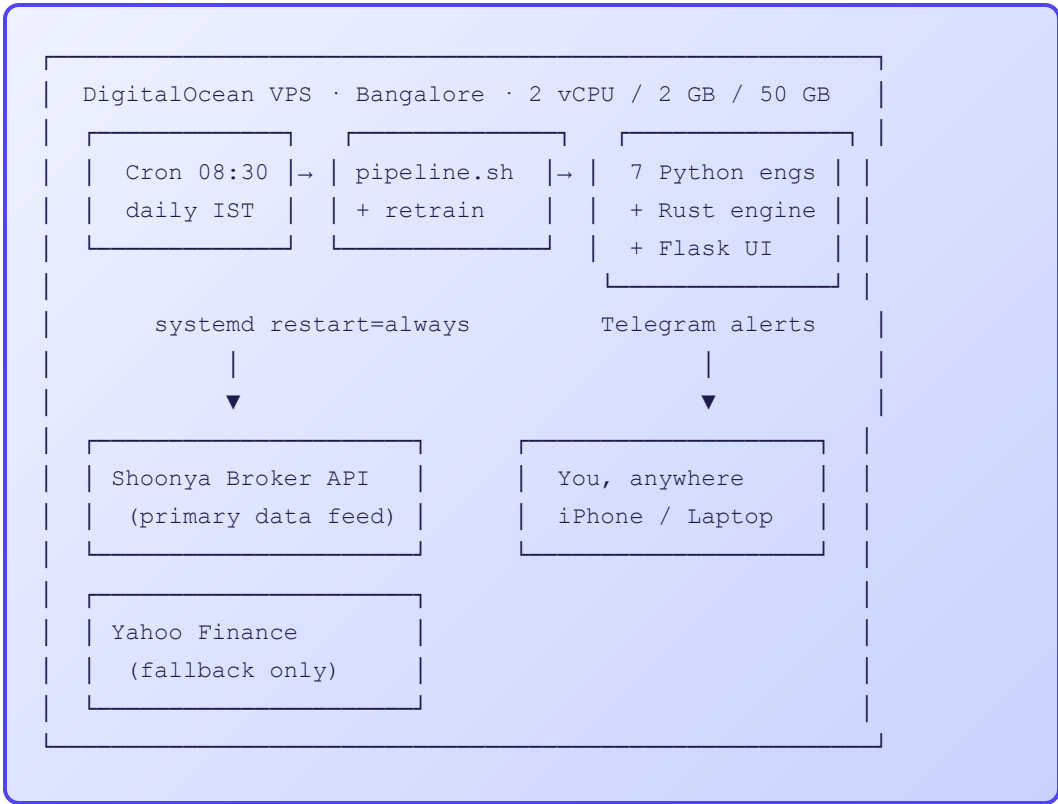
The repeating pattern

Apr 17: Laptop slept in bag while markets were open — all 5 paper-trade engines died.

Apr 20: Yahoo returned HTTP 429 during morning retrain — 10 minutes of scrambling before market open.

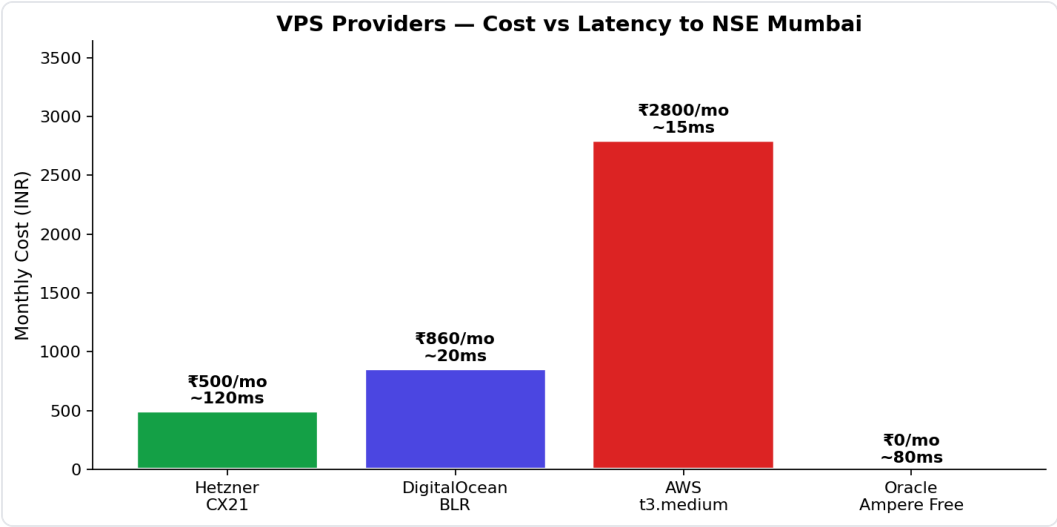
These are not bugs in our code. They are fundamental limits of the platform: **a laptop is the wrong host for a 24/7 trading system.**

Target Architecture



6 pain points are either eliminated or dramatically reduced by the cloud move

VPS Provider Comparison



Cost vs latency to NSE Mumbai. Bangalore-hosted DigitalOcean is the sweet spot.

Provider	Specs	Mumbai Latency	Cost/mo	Notes
Hetzner CX21	3 vCPU, 4 GB	~120 ms	₹500	EU-based. Great value, higher latency hurts intraday data refresh
DigitalOcean BLR ← pick	2 vCPU, 2 GB	<20 ms	₹860	Bangalore. Right abstraction level. Simple UX. Predictable pricing
AWS t3.medium	2 vCPU, 4 GB	<15 ms	₹2,800	Mumbai region. 3× cost for complexity you don't need
Oracle Ampere Free	4 vCPU, 24 GB	~80 ms	₹0	Actually free. ARM arch (Rust rebuild needed). Risk: account cancellation

Why DigitalOcean wins

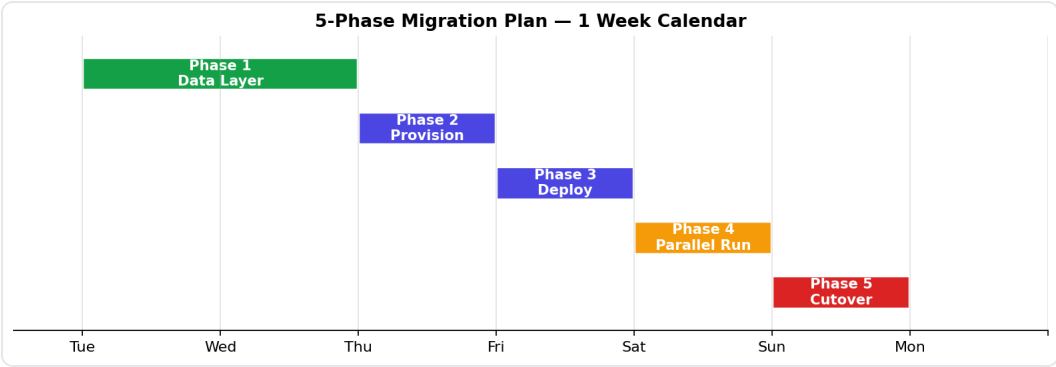
Latency: <20 ms to NSE means Yahoo/Shoonya calls return in under half a second — critical during the 9:15-9:20 opening volatility.

Cost predictability: Fixed ₹860/mo. No AWS surprise bills from bandwidth or API calls.

Simplicity: 5-minute droplet provisioning. One-click snapshots. Clean dashboard. You won't need to hire a DevOps.

SSH-first: Same workflow as laptop — git pull, systemd restart, done.

5-Phase Plan – 1 Week



One week from green-light to cutover. Phase 4 (Fri) is parallel-run — zero-risk verification.

Phase 1

Tuesday – Wednesday

Data Layer Fix — Shoonya Primary, Yahoo Fallback

Modify `prototype/v4/composite_scorer.py` to prefer Shoonya WebSocket when available, fall back to `yfinance` if Shoonya disconnects. Existing `scripts/shoonya-download.py` already proves connectivity works. Add retry-with-exponential-backoff for Yahoo calls.

Risk: Low. Shoonya integration is isolated. Yahoo stays as safety net.

Phase 2

Wednesday

Provision VPS

Sign up DigitalOcean. Create Bangalore droplet (2 vCPU / 2 GB / 50 GB, Ubuntu 24.04). SSH key auth (no passwords). UFW firewall (allow only 22 + 5050 + 8080 + Cloudflare tunnel). Fail2ban. Install Python 3.11, Rust, PostgreSQL 15, systemd.

Risk: Low. Standard provisioning. ~2 hours total.

Phase 3

Thursday

Code Deploy + Systemd Services

`git clone` repo. `rsync` ML models (`prototype/v4/models/archive/*`) + CSV data. `cargo build --release` Rust engine. Write systemd unit files for each engine (Restart=always, User=tradepilot, EnvironmentFile=.env). Enable cron for `run-v5-tomorrow.sh` at 08:30 IST.

Risk: Medium. Rust compile takes ~4 min. Systemd units need careful dependency ordering.

Phase 4

Friday

Parallel Run

Both laptop AND VPS run the same market day. Telegram alerts from both distinguish source. At EOD, compare P&L engine-by-engine — should be identical. If P&L diverges >1%, investigate.

Risk: Low. Paper trading only, no real money. Divergence is diagnostic, not damaging.

Phase 5

Following Monday

Cutover

Laptop engines stopped permanently. VPS is primary. Laptop kept as cold backup for 2 weeks (can restart for emergency recovery). Monitor VPS Telegram feed for first week — if uptime stays above 99%, decommission laptop completely.

Risk: Low. Rollback is just restarting laptop. No data loss — position state syncs nightly.

Key Design Decisions

1. Data source

Option	Reliability	Cost	Verdict
Yahoo Finance (current)	Low — 429 rate limits	Free	Fallback only
Shoonya broker API ← pick	High — official NSE feed	Free w/ account	Primary
Paid vendor (Truedata, etc)	Highest	₹2000+ /mo	Overkill for paper trading

2. Storage

Keep position state in JSON files (simple, atomic writes already applied in learning #004). DevPilot PostgreSQL stays on VPS for analytics. Nightly `pg_dump` to a backup disk. ML model archive (`prototype/v4/models/archive/YYYY-MM-DD/`) remains the rollback truth.

3. Process manager

Replace `nohup + custom watchdog` with `systemd`. Each engine is a `.service` file with `Restart=always`. `Systemd` handles: crash-restart, log rotation, startup ordering, dependency graph, user/group isolation. Zero custom monitoring code needed.

Why not Docker?

Tempting, but unnecessary complexity for this workload. Docker adds ~300 MB per container, image rebuild per deploy, and extra layer of debugging. Native `systemd` with Python `venv` is simpler for 7-process workload. Reconsider when we add horizontal scaling or multi-tenant.

4. Remote access

Three channels:

Channel	Use case	Setup
SSH	Live debugging, deploys	SSH keys on iPhone/laptop, 5 min
Flask dashboard	Mobile viewing of status/P&L	Cloudflare Tunnel → free subdomain
Telegram	Push alerts	Already working

5. Secrets

`.env` file on VPS with `0600` permissions. Contains Telegram bot token, Shoonya API creds, DB password. Never committed to git. systemd loads via

`EnvironmentFile=/home/tradepilot/.env` . Rotate Shoonya password quarterly.

Before / After Pain Matrix

Problem today	Root cause	Resolution on VPS
Laptop sleep kills engines	macOS clamshell firmware event	VPS runs 24/7. No physical lid.
Yahoo HTTP 429 during morning retrain	Laptop IP shared with other yfinance users	Shoonya official feed. Different host, different IP if needed.
Manual 08:30 IST start every day	No scheduling on laptop	systemd timer fires automatically. You sleep in.
Lid-open-in-bag gymnastics	No VPS to run on	Laptop freed for coding. No gymnastics.
Home WiFi drop kills live session	Single network dependency	Data center uplink 99.99%. Your WiFi is just for dashboard viewing.
Location lock — only works from Mumbai desk	Physical machine	SSH from phone anywhere. Push alerts reach you in meetings.
Battery life / heat stress	Laptop isn't designed for 6-hour CPU sessions	VPS has proper cooling and power. Laptop gets its life back.

Cost-Benefit

Annual math

- Direct cost:** ₹860/mo × 12 = **₹10,320/year** for the VPS.
- Data:** Shoonya is ₹0. Yahoo as fallback is ₹0.
- Time savings:** 10 min/day of manual pre-flight × 250 trading days = 41 hours of your life back.
- Trade loss prevention:** Apr 17's laptop-sleep event lost ~Rs 500 of realized wins. One prevented incident pays for 7 months of VPS.
- Psychological:** No more bag-gymnastics. No more pre-market anxiety. Priceless.

What You Need To Decide

Decision	Options	Default
Provider	DigitalOcean BLR / Hetzner / AWS / Oracle	DigitalOcean BLR ₹860/mo
Shoonya API access	Use existing account / Open new / Skip for now	Use existing (you have one)
Go-Live day	Next Monday / Delay by 1 week / Don't migrate	Next Monday (Apr 27)
Keep laptop as backup?	2 weeks / 1 month / Decommission immediately	2 weeks cold backup
Paid data vendor?	None (Shoonya) / Truedata / TrueBars	None — revisit after 3 months

Next Concrete Steps

1. **Today:** Approve or modify this plan
2. **Tomorrow:** Sign up DigitalOcean · Get Shoonya API creds from your broker
3. **Wednesday:** I spin up the VPS, run Phase 1 Shoonya integration locally first
4. **Thursday-Friday:** Phase 3 deploy + Phase 4 parallel run
5. **Next Monday:** Cutover. VPS is primary. You sleep in on Tuesday.

Stop Fighting the Host.

The laptop is a great dev machine. It is not a trading platform.
Every hour spent keeping it alive — caffeinate hacks, lid-open bags,
sudo pmset incantations — is an hour not spent on strategy.

*"The best trading infrastructure is the one you stop
noticing."*

Approve the plan. One week later, you'll never touch run-v5-
tomorrow.sh on your laptop again.

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